RRC-Raptor-V4H DDR

- User's Manual: Software

V0.0.2

Contents

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i. VERSION LIST

Description	Version	Date	Author
RTX_DDR	V0.0.1	2024/06/26	Jerry
Translate the content into English.	V0.0.2	2024/11/20	Jerry

ii. HASHTAG

#V4H
#DDR
#BSP
#ubuntu

1. Overview

Target：Introduce how to check the DDR size and use the stressapptest tool to test DDR.

2. Check DDR size

You can check the DDR size using the free command. As shown in Figure 1, the value 7304 represents 7304 megabytes, which indicates that the DDR size is 8GB.

```
root@v4x:~# free -m
              total        used        free      shared  buff/cache   available
Mem:           7304         618        6605          19          80        6546
Swap:             0           0           0
```

Figure 1. Example

3. Test DDR

You can use stressapptest to test the DDR performance during data access.

The following is an introduction to commonly used stressapptest parameters.

-s seconds : Number of seconds to run
-M mbytes : Megabytes of ram to test
-m threads : Number of memory copy threads to run
-C threads : Number of memory CPU stress threads to run.
-W : Use more CPU-stressful memory copy

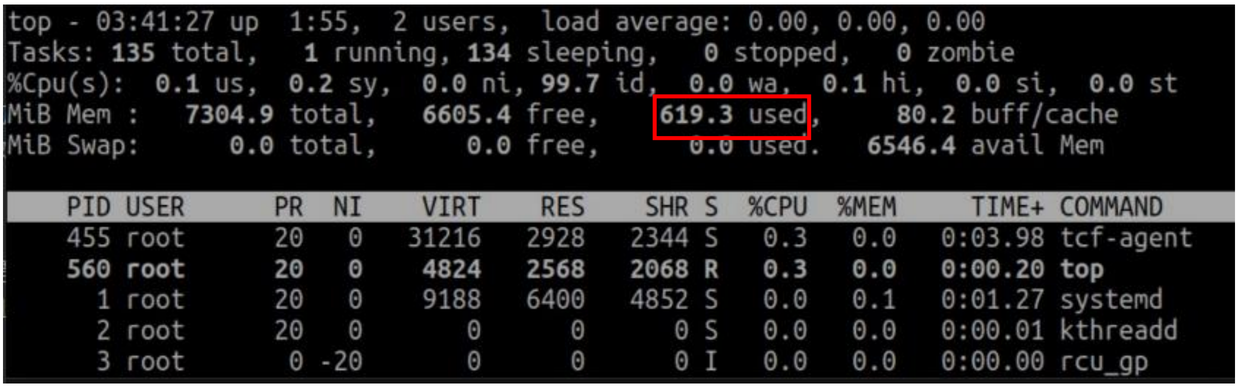
So, if we want to test a 4GB DDR for 1 hour, we can use the following command to perform the test.

stressapptest -s 3600 -M 4096 -m 4 –W

Users can also adjust the parameters according to their own needs.

The following are the actual test conditions.

1. Before testing, use the top command to check the current system memory usage. Currently, only 619 megabytes are in use..



```
top - 03:41:27 up  1:55,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 135 total,   1 running, 134 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.2 sy,  0.0 ni, 99.7 id,  0.0 wa,  0.1 hi,  0.0 si,  0.0 st
MiB Mem :   7304.9 total,   6605.4 free,    619.3 used,     80.2 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   6546.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    455 root      20   0   31216   2928   2344 S   0.3   0.0   0:03.98 tcf-agent
    560 root      20   0    4824   2568   2068 R   0.3   0.0   0:00.20 top
      1 root      20   0    9188   6400   4852 S   0.0   0.1   0:01.27 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
```

Figure 2. Memory status before Test

2. After running stressapptest, it has increased to 4722 MB in use, indicating that stressapptest is testing this memory..

```
root@v4x:~# stressapptest -s 3600 -M 4096 -m 4 -W
2024/06/26-03:48:56(UTC) Log: Commandline - stressapptest -s 3600 -M 4096 -m 4 -W
2024/06/26-03:48:56(UTC) Stats: SAT revision 1.0.9_autoconf, 64 bit binary
2024/06/26-03:48:56(UTC) Log:  @ jerry-pc on Tue Jun 25 05:27:46 UTC 2024 from open source release
2024/06/26-03:48:56(UTC) Log: 1 nodes, 4 cpus.
2024/06/26-03:48:56(UTC) Log: Prefer plain malloc memory allocation.
2024/06/26-03:48:56(UTC) Log: Using mmap() allocation at 0xfffea4000000.
2024/06/26-03:48:56(UTC) Stats: Starting SAT, 4096M, 3600 seconds
2024/06/26-03:48:58(UTC) Log: region number 39 exceeds region count 1
2024/06/26-03:48:58(UTC) Log: Region mask: 0x1
2024/06/26-03:49:08(UTC) Log: Seconds remaining: 3590

jerry@jerry: ~ 98x15
top - 03:49:14 up  2:03,  2 users,  load average: 0.89, 0.20, 0.06
Tasks: 136 total,   1 running, 135 sleeping,   0 stopped,   0 zombie
%Cpu(s): 99.2 us,  0.4 sy,  0.0 ni,  0.0 id,  0.0 wa,  0.3 hi,  0.1 si,  0.0 st
MiB Mem :   7304.9 total,   2501.8 free,   4722.9 used,     80.2 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   2442.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  562 root      20   0 4838904   4.0g   2588 S 398.3  56.1   1:09.47 stressapptest
  560 root      20   0    4824   2568   2068 R   0.3   0.0   0:01.98 top
    1 root      20   0    9188   6400   4852 S   0.0   0.1   0:01.28 systemd
    2 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kthreadd
    3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root      20   0       0      0      0 I   0.0   0.0   0:00.10 kworker/0:0-events
    8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
```

Figure 3 Memory status during test

3. Finally, you can check the ending message to see if any errors occurred during the test..

```
2024/06/26-03:55:59(UTC) Stats: Found 0 hardware incidents
2024/06/26-03:55:59(UTC) Stats: Completed: 357004.00M in 60.01s 5949.43MB/s, with 0 hardware incidents, 0 errors
2024/06/26-03:55:59(UTC) Stats: Memory Copy: 357004.00M at 5949.80MB/s
2024/06/26-03:55:59(UTC) Stats: File Copy: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Stats: Net Copy: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Stats: Data Check: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Stats: Invert Data: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Stats: Disk: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC)
2024/06/26-03:55:59(UTC) Status: PASS - please verify no corrected errors
2024/06/26-03:55:59(UTC)
```

Figure 4 Memory test result

4. APPENDIX